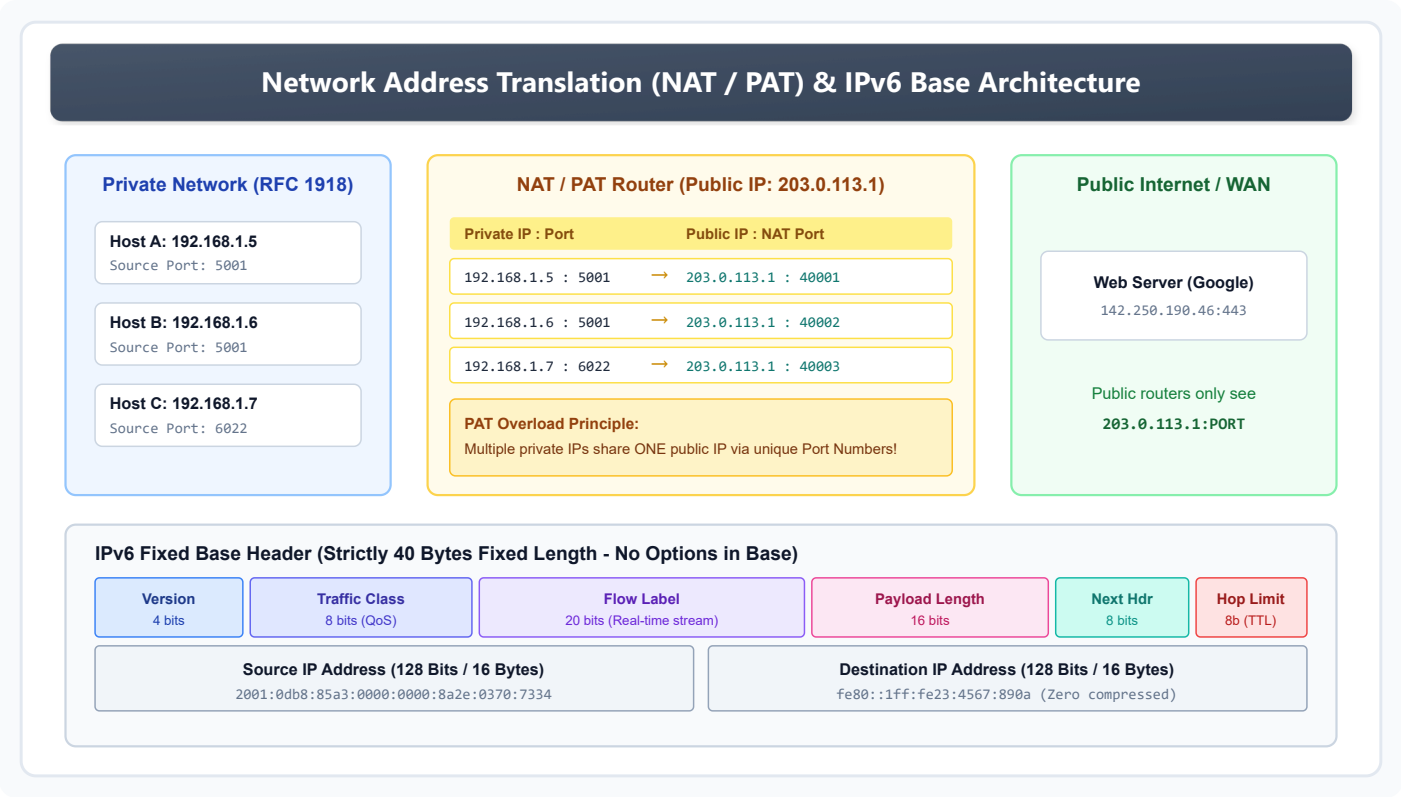


Module 07: Network Address Translation (NAT), PAT, and IPv6 Architecture

Unit 3: Network Layer NAT/PAT & IPv6 Evolution

Private-to-Public IP Translation, Port Address Translation (PAT) Overload, IPv6 128-bit Addressing Architecture, 40-Byte Fixed Base Header, and Transition Strategies.



1. NAT Translation Types

- Static NAT: 1-to-1 permanent private-to-public mapping.
- Dynamic NAT: Pool of public IPs assigned on first-come first-served basis.
- PAT (Port Address Translation / NAT Overload): Thousands of private hosts share 1 single public IP using unique L4 Port numbers!

2. IPv4 vs IPv6 Architectural Comparison

Feature	IPv4	IPv6
Address Length	32 bits (4 bytes)	128 bits (16 bytes)
Header Size	Variable: 20 to 60 Bytes	Fixed: Strictly 40 Bytes
Header Checksum	Present (recomputed at every hop)	Eliminated (handled by L2 & L4)
Fragmentation	Routers & Source host	Source host only (Path MTU Discovery)
Broadcast	Supported (Limited & Directed)	Eliminated (Replaced by Multicast/Anycast)